

ECE 3337: Test #2

Fall 2023

Last name of student POSTED	Student ID number SOLUTIONS
First name of student _____	Email address _____

Do This First:

1. Make sure that you have all the pages of the exam.
2. Fill in the above information.
3. Turn off and put away all electronic gadgets (laptop, tablet, smartphone, calculator, smart watch, ...)

Exam Rules:

1. You are allowed to bring one 8.5" x 11" handwritten crib sheet (double sided).
2. Indicate your answers in the spaces provided. Use the back of the page to carry out calculations.

Q1	Q2	Q3	Q4	Q5	Q6	Total
20	20	25	15	10	10	100
Laplace Transform	Inverse Laplace Transform	Circuit Analysis	Response Partitions	Op Amp Circuits	Block Diagrams	_____
_____	_____	_____	_____	_____	_____	_____

1. (20 points) Laplace Transforms

a. (6 points) Find $X(s)$, the Laplace transform of the signal $x(t) = \delta(2t - 1) + r(t - 1) + t^3 e^{-t} u(t)$.

$$\mathcal{L}[\delta(2t - 1)] = \mathcal{L}\left[\frac{1}{2}\delta\left(t - \frac{1}{2}\right)\right] = \frac{1}{2}e^{-\frac{s}{2}} \quad \textcircled{A}$$

$$\mathcal{L}[r(t - 1)] = e^{-s}\mathcal{L}[r(t)] = e^{-s}/s^2 \quad \textcircled{B}$$

$$\mathcal{L}[t^3 e^{-t} u(t)] = \frac{3!}{(s+1)^4} \quad \textcircled{C}$$

$$X(s) = \frac{1}{2}e^{-s/2} + \frac{e^{-s}}{s^2} + \frac{6}{(s+1)^4}$$

Grading:

- +2 pts for each part $\textcircled{A}$, $\textcircled{B}$, $\textcircled{C}$
- minimum 1 pt for trying

(b) (6 points) Given a signal $x_2(t)$, find the Laplace transform of $y(t) = 3e^{-2t}x_2(2t - 5)$ in terms of $X_2(s)$

$$\mathcal{L}[x_2(2t)] = \frac{1}{2}X_2\left(\frac{s}{2}\right) \quad \textcircled{A}$$

$$\mathcal{L}[x_2(2t - 5)] = \mathcal{L}\left[x_2\left(2\left(t - \frac{5}{2}\right)\right)\right] = \frac{1}{2}e^{-\frac{5}{2}s}X_2\left(\frac{s}{2}\right) \quad \textcircled{B}$$

$$\mathcal{L}[e^{-2t}x_2(2t - 5)] = \frac{1}{2}e^{-\frac{s}{2}(s+2)}X_2\left(\frac{(s+2)}{2}\right) \quad \textcircled{C}$$

$$\Rightarrow Y(s) = \frac{3}{2}e^{-\frac{s}{2}(s+2)}X_2\left(\frac{(s+2)}{2}\right)$$

Grading:

- 2 pts for each part $\textcircled{A}$, $\textcircled{B}$, $\textcircled{C}$
- min. 1 pt for trying

c. (8 points) Find the Laplace transform of $z(t) = t e^{-t} \sin(t) u(t)$, and find the values of all its poles and zeros.

$$\mathcal{L}[\sin(t) u(t)] = \frac{1}{s^2 + 1} \quad \textcircled{A}$$

$$\mathcal{L}[e^{-t} \sin(t) u(t)] = \frac{1}{(s+1)^2 + 1} \quad \textcircled{B}$$

$$\mathcal{L}[t e^{-t} \sin(t) u(t)] = -\frac{d}{ds} \left\{ \frac{1}{(s+1)^2 + 1} \right\} \quad \textcircled{C}$$

$$= \frac{2(s+1)}{((s+1)^2 + 1)^2}$$

$s = -1$ zero
 $s = -1 \pm j$ poles repeated } $\textcircled{D}$

Grading:

- 2 pts for trying
- + 2 pts each $\textcircled{A}, \textcircled{B}, \textcircled{C}$
- Full points for correct answer $\textcircled{D}$

2. (20 points) Inverse Laplace Transform

Find the inverse Laplace transforms for the following systems.

a. (10 points) $X(s) = \frac{e^{-3s}(s+7)}{(s^2+10s+29)}$

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$$\begin{aligned} X(s) &= e^{-3s} \frac{(s+7)}{(s^2+10s+29)} \\ &= e^{-3s} \frac{s+5+2}{s^2+2 \cdot 5 \cdot s+25+4} \\ &= e^{-3s} \frac{(s+5)+2}{(s+5)^2+2^2} \\ &= e^{-3s} \left\{ \frac{(s+5)}{(s+5)^2+2^2} + \frac{2}{(s+5)^2+2^2} \right\} \end{aligned}$$

From Table 3-2 #14 and #13,

$$\frac{(s+5)}{(s+5)^2+2^2} \xleftrightarrow{\mathcal{L}^{-1}} e^{-5t} \cos(2t) u(t) \text{ and } \frac{2}{(s+5)^2+2^2} \xleftrightarrow{\mathcal{L}^{-1}} e^{-5t} \sin(2t) u(t)$$

Also using the time shifting property from Table 3-1 #4, the final answer is

$$x(t) = [e^{-5(t-3)} \cos(2(t-3)) + e^{-5(t-3)} \sin(2(t-3))] u(t-3)$$

$$x(t) = [\cos(2(t-3)) + \sin(2(t-3))] e^{-5(t-3)} u(t-3)$$

b. (10 points)

2 points for trying

$$\begin{aligned} X(s) &= \frac{3s}{(s+1)^3} \\ &= 3 \frac{(s+1-1)}{(s+1)^3} = 3 \frac{(s+1)}{(s+1)^3} - \frac{3}{(s+1)^3} \\ &= \frac{3}{(s+1)^2} - \frac{3}{(s+1)^3} \\ &= 3 \frac{1}{(s+1)^2} - \frac{3}{2} \frac{2}{(s+1)^3} \end{aligned}$$

Using Table 3-2 #4 and #5 and the frequency shifting property #5 from Table 3-1

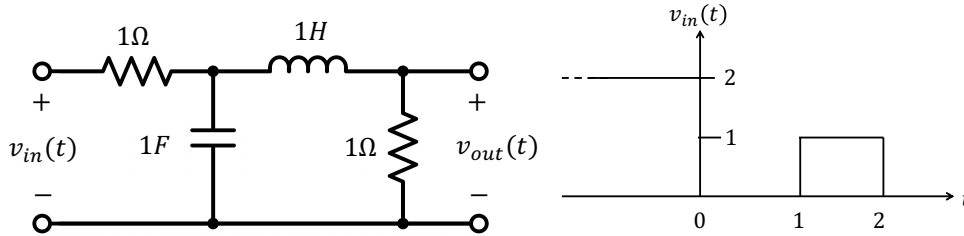
$$\frac{1}{(s+1)^2} \xleftrightarrow{\mathcal{L}^{-1}} e^{-t} t u(t) \text{ and } \frac{2}{(s+1)^3} \xleftrightarrow{\mathcal{L}^{-1}} e^{-t} t^2 u(t)$$

Therefore, the final answer is

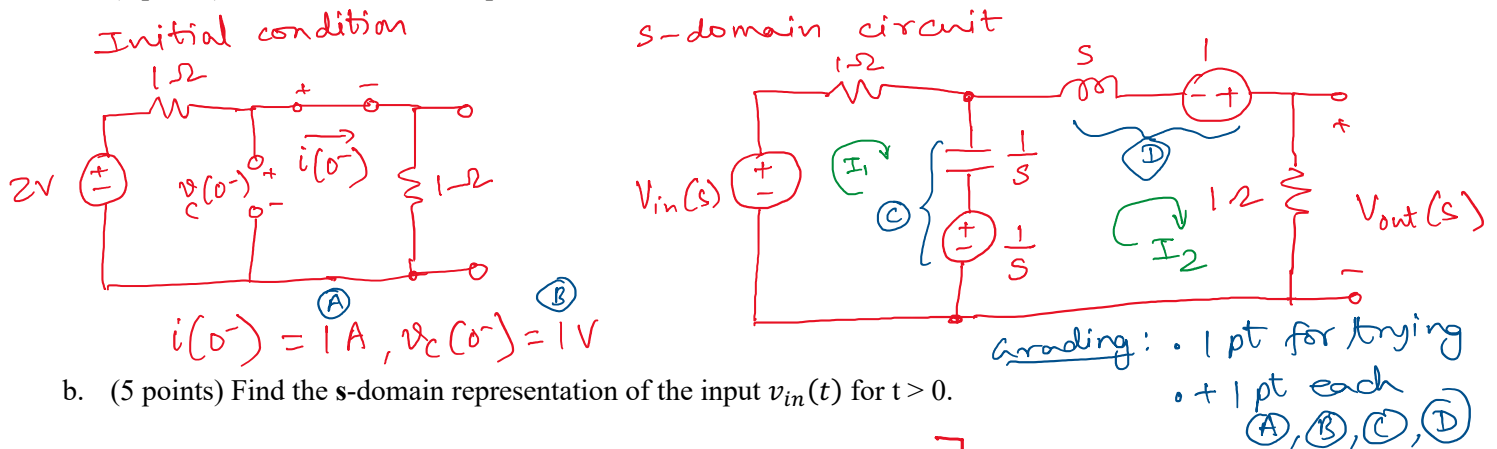
$$x(t) = 3e^{-t} t u(t) - \frac{3}{2} e^{-t} t^2 u(t) = [3t - \frac{3}{2} t^2] e^{-t} u(t)$$

3. (25 points) Circuit Analysis

For the circuit below, the input $V_{in}(t) = x(t)$ is given in the graph. The units for $x(t)$ is Volts [V].



- a. (5 points) Draw the s-domain representation of the circuit with the initial conditions indicated.



- b. (5 points) Find the s-domain representation of the input $v_{in}(t)$ for $t > 0$.

$$V_{in}(s) = \mathcal{L}[u(t-1) - u(t-2)]$$

$$= \frac{e^{-s}}{s} - \frac{e^{-2s}}{s} = \frac{1}{s} [e^{-s} - e^{-2s}]$$

Grading: • 1 pt for trying
• +2 pts each (A), (B)

- c. (10 points) Write the mesh current equations in the s-domain needed to find the output $V_{out}(s)$. Express your answer by filling in the entries in the Cramer's method matrices shown below. Here, I_1 and I_2 represent the mesh currents (left to right), and V_1 , V_2 represent mesh voltages.

$$\begin{pmatrix} A_1 & A_2 \\ A_3 & A_4 \end{pmatrix} \begin{pmatrix} I_1 \\ I_2 \end{pmatrix} = \begin{pmatrix} V_1 \\ V_2 \end{pmatrix}$$

mesh equations

$$\left. \begin{aligned} I_1 \left(1 + \frac{1}{s} \right) + I_2 \left(-\frac{1}{s} \right) &= V_{in}(s) - \frac{1}{s} \\ I_1 \left(-\frac{1}{s} \right) + I_2 \left(\frac{1}{s} + s + 1 \right) &= \frac{1}{s} + 1 \end{aligned} \right\}$$

Grading: • 2 pts trying
• +1 pt each A_1, A_2, A_3, A_4
 V_1, V_2
• +2 pts correct I_2

$$\begin{pmatrix} 1 + \frac{1}{s} & -\frac{1}{s} \\ -\frac{1}{s} & \frac{1}{s} + s + 1 \end{pmatrix} \begin{pmatrix} I_1 \\ I_2 \end{pmatrix} = \begin{pmatrix} V_{in}(s) - \frac{1}{s} \\ \frac{1}{s} + 1 \end{pmatrix}$$

$$\Delta = A_1 A_4 - A_3 A_2 = s + \frac{2}{s} + 2 = \frac{s^2 + 2s + 2}{s}$$

no need for
 ΔI_1

$$\Delta I_2 = \left(1 + \frac{1}{s}\right) \left(\frac{1}{s} + 1\right) - \left(-\frac{1}{s}\right) \left(V_{in} - \frac{1}{s}\right) = \frac{V_{in}(s) + s + 2}{s}$$

$$\Rightarrow I_2(s) = \frac{V_{in}(s) + s + 2}{s^2 + 2s + 2}$$

d. (5 points) Write the formula for $V_{out}(s)$. Your answer should be in the form of a ratio of polynomials in s .

$$V_{out}(s) = I_2(s) \cdot 1 = \frac{V_{in}(s) + s + 2}{s^2 + 2s + 2}$$

Grading:

- 1 pt trying
- 2 pts partially correct
- 5 pts correct answer

$$= \frac{\frac{1}{s} [e^{-s} - e^{-2s}] + s + 2}{s^2 + 2s + 2}$$

$$= \frac{e^{-s} - e^{-2s} + s^2 + 2s}{s(s^2 + 2s + 2)} //$$

4. (15 points) System Response Partitions

A linear time-invariant system is described by the following LCCDE

$$\frac{d^2 y(t)}{dt^2} + 4 \frac{dy(t)}{dt} + 3y(t) = 2x(t)$$

The input is $x(t) = e^{-2t}u(t)$, with the initial conditions $y(0^-) = 1$, $y'(0^-) = 2$.

- a. (5 points) Derive a formula for the output $\mathbf{Y(s)}$. Your answer should be expressed as a ratio of polynomials with a fully factored denominator.

$$y(t) \xleftrightarrow{\mathcal{L}} \mathbf{Y(s)}$$

1 point for trying

$$\frac{dy(t)}{dt} \xleftrightarrow{\mathcal{L}} s\mathbf{Y(s)} - y(0^-) = \underline{s\mathbf{Y(s)} - 1} \quad \textcircled{A}$$

$$\frac{d^2 y(t)}{dt^2} \xleftrightarrow{\mathcal{L}} s^2\mathbf{Y(s)} - sy(0^-) - y'(0^-) = \underline{s^2\mathbf{Y(s)} - s - 2} \quad \textcircled{B}$$

The LCCDE transforms to:

$$s^2\mathbf{Y(s)} - s - 2 + 4[s\mathbf{Y(s)} - 1] + 3\mathbf{Y(s)} = 2\mathbf{X(s)}$$

$$\mathbf{Y(s)}[s^2 + 4s + 3] - s - 6 = 2\mathbf{X(s)}$$

$$\mathbf{Y(s)} = \frac{2\mathbf{X(s)} + 7}{s^2 + 4s + 3} = \frac{2\mathbf{X(s)} + s + 6}{(s + 3)(s + 1)} \quad \textcircled{C}$$

1 point each
for $\textcircled{A}$ - $\textcircled{D}$

Since $x(t) = e^{-2t}u(t)$, $\mathbf{X(s)} = \frac{1}{s+2}$. Therefore,

$$\mathbf{Y(s)} = \frac{\frac{2}{s+2} + s + 6}{(s + 1)(s + 3)} = \frac{s^2 + 8s + 14}{(s + 1)(s + 2)(s + 3)} \quad \textcircled{D}$$

- b. (5 points) Calculate the forced response of this system in the Laplace domain.

1 point for trying

The partial fraction decomposition of

$$\mathbf{Y(s)} = \frac{s^2 + 8s + 14}{(s + 1)(s + 2)(s + 3)} = \frac{A}{(s + 1)} + \frac{B}{(s + 2)} + \frac{C}{(s + 3)} \quad \textcircled{A}$$

The modes of the output are a combination of the modes of the system and the modes of the input. The input $x(t) = e^{-2t}u(t)$ is associated with the $(s + 2)$ mode. The forced response corresponds to the output term involving the modes in the input. Here, we just need to find B in the above partial fraction expansion to find the forced response. $\textcircled{B}$

$$B = \mathbf{Y(s)}(s + 2)|_{s=-2} = -2 \quad \textcircled{C}$$

Therefore, the forced response is

$$-\frac{2}{(s + 2)} \quad \textcircled{D}$$

1 point for
 $\textcircled{A}$ - $\textcircled{D}$

c. (5 points) Calculate the final value $y(\infty)$ of this system's response.

1 point for trying

Using the final value theorem,

(A)

$$y(\infty) = \lim_{s \rightarrow 0} sY(s) \quad \text{--- (B)}$$

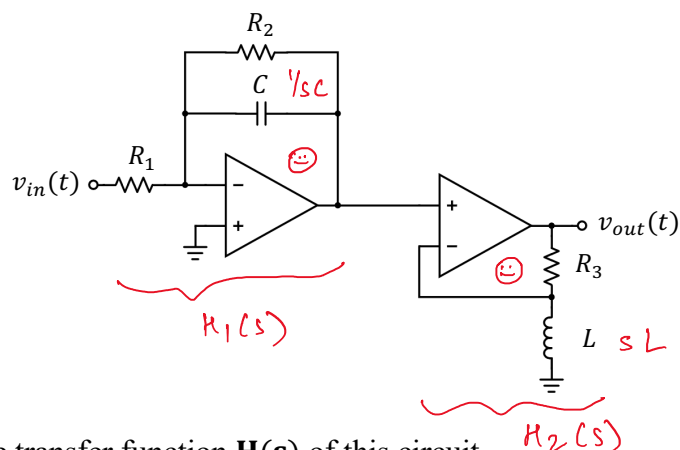
$$\lim_{s \rightarrow 0} sY(s) = \lim_{s \rightarrow 0} \frac{s(s^2 + 8s + 14)}{(s+1)(s+2)(s+3)} = 0 \quad \text{(D)}$$

(C)

1 point for (A) - (D)

5. (10 points) Operational Amplifiers

For the circuit below, there was no energy stored initially in the capacitor.



a. (6 points) Calculate the transfer function $\mathbf{H(s)}$ of this circuit.

$$H(s) = H_1(s) H_2(s)$$

$$H_1(s) = - \left(\frac{R_2}{R_1} \right) \frac{1}{(1 + R_2 C s)} \quad \textcircled{A}$$

$$H_2(s) = \frac{R_3 + sL}{sL} \quad \textcircled{B}$$

$$\Rightarrow H(s) = - \left(\frac{R_2}{R_1} \right) \cdot \frac{(R_3 + sL)}{(sL)(1 + R_2 C s)} \quad \textcircled{C}$$

Grading

- 1 pt for trying
- 2 pts each $\textcircled{A}, \textcircled{B}$
- +1 pt $\textcircled{C}$

b. (4 points) Identify the poles and zeros of the transfer function $\mathbf{H(s)}$.

poles: $s = 0 \quad \textcircled{A}, \quad s = -\frac{1}{R_2 C} \quad \textcircled{B}$

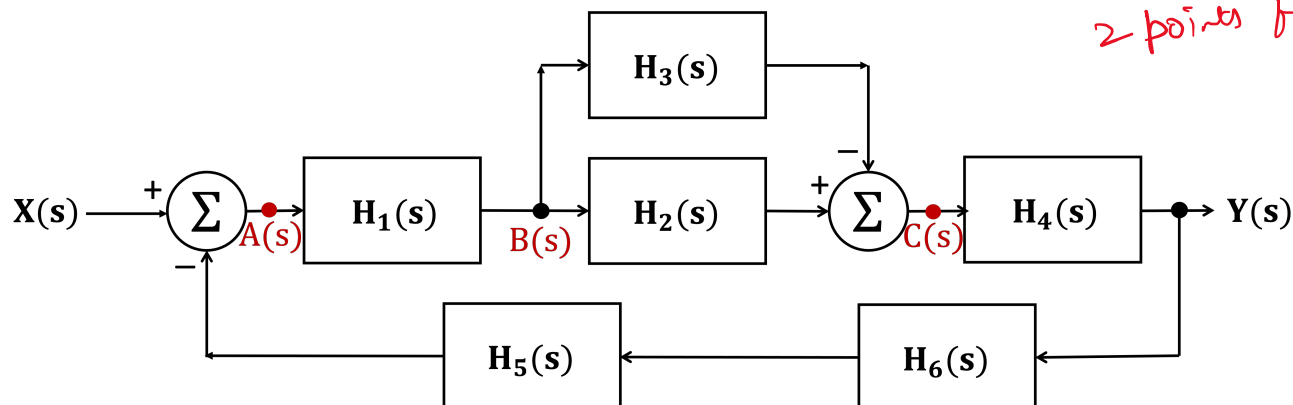
zero: $s = -\frac{R_3}{L} \quad \textcircled{C}$

Grading

- 1 pt trying
- +1 pt each for $\textcircled{A}, \textcircled{B}, \textcircled{C}$

6. (10 points) System Block Diagrams

Calculate the overall transfer function $\mathbf{H(s)}$ for the following connection of LTI systems. Your final answer should be expressed in the Laplace domain.



Assuming an s dependence in all functions.

$$A = X - Y \cdot (H_6 \cdot H_5) \quad \text{--- (A)}$$

$$B = A \cdot H_1 \quad \text{--- (B)}$$

$$C = B \cdot (H_2 - H_3) \quad \text{--- (C)}$$

$$Y = C \cdot H_4 \quad \text{--- (D)}$$

$$C = B \cdot (H_2 - H_3) = A \cdot H_1 \cdot (H_2 - H_3) = [X - Y \cdot (H_6 \cdot H_5)] \cdot H_1 \cdot (H_2 - H_3)$$

$$Y = C \cdot H_4 = [X - Y \cdot (H_6 \cdot H_5)] \cdot H_1 \cdot (H_2 - H_3) \cdot H_4 \quad \text{--- (E)}$$

$$Y \cdot [1 + H_6 \cdot H_5 \cdot H_1 \cdot (H_2 - H_3) \cdot H_4] = X \cdot [H_1 \cdot (H_2 - H_3) \cdot H_4] \quad \text{--- (F)}$$

$$Y = \frac{X \cdot [H_1 \cdot (H_2 - H_3) \cdot H_4]}{1 + H_6 \cdot H_5 \cdot H_1 \cdot (H_2 - H_3) \cdot H_4}$$

The overall transfer function is

$$H = \frac{Y}{X} = \frac{[H_1 \cdot (H_2 - H_3) \cdot H_4]}{1 + H_6 \cdot H_5 \cdot [H_1 \cdot (H_2 - H_3) \cdot H_4]} \quad \text{--- (G)}$$

1 point each for (A)-(F)
2 points for (G)